

Element C: Presentation and Justification of Solution Design Requirements

Many of the teachers and students at our school are wasting paper. They are not recycling or using the paper for other projects. This is a pretty big issue as it affects the environment. For this project, our team is trying to find a way to reuse a stack of used paper so that it can be used again, otherwise it would just be thrown in the trash, which is not very eco-friendly. We are developing the solution for our school. Our design requirements are listed from highest priority to lowest.

- The process to reuse the paper must be safe, including no sharp objects or edges, and the final product must be safe to touch. This is a high priority requirement.
- Reuse at least 90% of the paper. This is a high priority requirement.
- One side of each piece of paper should be blank. This is a high priority requirement.
- The paper should be connected in some way. This is a high priority requirement.
- The paper sheets need to be 8.5 inches by 11 inches. This is a medium priority requirement.
- We should stay within a \$2.00 per solution budget. This is a low priority requirement.
- We must use at least 50 sheets of paper. This is a low priority requirement.

.For safety we are using a paper guillotine and we found an article that takes us through the steps on using one (*How to Properly Use a Guillotine Paper Trimmer*,

2017). We did research on picking a glue that will hold our paper together (*Guide*, n.d.). All of these design requirements are necessary so that our stakeholders will be satisfied with using the repurposed paper. One of the teachers Mr. K told us to expand on the design requirements. One side of the paper must be blank. He told us that the non-blank side must not be distracting. This change will make it more likely for teachers and students to use. Our design requirements will help us build a better product for our stakeholders, and it will be easier for our product to be used by our stakeholders. Seeing as how many of our stakeholders already use something similar to our final solution, we believe that our products will be quickly adopted by our stakeholders, as they are better for the environment. Our goal is that our school will benefit from the scrap paper and it will be better for the environment.

References:

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